

# Omer Kilic

[linkedin.com/in/omer-kilic16](https://www.linkedin.com/in/omer-kilic16) · [github.com/omerkilicdev](https://github.com/omerkilicdev) · [omerkilic.github.io](https://omerkilic.github.io)

16omerkilic@gmail.com · (551) 775-3655 · North Jersey, NJ

Aerospace Engineering · Computational Aerodynamics · Scientific Computing · Manufacturing & Testing · Propulsion

## EDUCATION

**Rutgers University** · New Brunswick, NJ · B.S. Aerospace Engineering Expected May 2028  
GPA 4.00/4.00 · Aresty Research Scholar · Richard Peters & Cecilia Santiago Peters Endowed Scholar

**Bergen Community College** · Paramus, NJ · A.S. Engineering Science 2025  
GPA 3.90/4.00 · Phi Theta Kappa · USDA 3SP Scholar (\$3,000)

**Bilkent University** · Ankara, Turkiye 2018 – 2020  
Economics coursework · Full scholarship covering tuition and stipend · National University Entrance rank: **1,085 of 2.5M** candidates

## TECHNICAL SKILLS

**Aerodynamics & Numerics** Fast Multipole Method, discrete-vortex methods, unsteady wake modeling, multipole expansions, 6-DOF store-separation trajectory analysis, flight dynamics and stability derivatives, CFD meshing and post-processing

**Scientific Computing** Julia, Python (NumPy, Pandas, Matplotlib, Jupyter), MATLAB, C++, Git, LaTeX/Typst

**Optimization & Verification** automatic differentiation, gradient-based design optimization, Monte Carlo / GCMC, surrogate and reduced-order modeling, numerical verification, error analysis and convergence study

**Design & Manufacturing** SolidWorks, composites manufacturing, rotor/propeller balancing, GD&T, NX/Fibersim workflow exposure, NCR/MRB

**Test & Instrumentation** wind-tunnel test exposure, sensor monitoring and diagnostics, CAN Bus, NEC code compliance

## RESEARCH & AEROSPACE EXPERIENCE

**Aresty Research Assistant** · Flapping-Wing Aerodynamics Lab, Rutgers MAE 2026 – Present  
PI: Prof. Mitsunori Denda · Bio-Inspired Micro Aerial Vehicles · [insectflight.rutgers.edu](https://insectflight.rutgers.edu)

- Built a Fast Multipole Method in Julia for a discrete-vortex flapping-wing solver and integrated it into the group's MATLAB 3-D wake code as a replacement velocity kernel; with vectorized exact evaluation, cut a 15-step simulation from **1,783 s to 50 s (35x)**.
- Derived multipole expansions for vortex rings and polygonal panels and built the full FMM pipeline with a priori far-field error bounds, verified against direct summation across **86 automated test groups**.
- Made the solver differentiable through automatic differentiation for gradient-based flapping-stroke design, and characterized the horizon over which gradients remain reliable.
- Authored a 10-part FMM tutorial series and group presentations; built and maintain the lab's hub, [flapping-wings.vercel.app](https://flapping-wings.vercel.app).

**Engineering Intern** · Turkish Aerospace Industries (TUSAŞ), Ankara Aug – Sep 2026  
Türkiye Internship Program (YTB) · selected via technical interview · UAV Manufacturing & KAAN Aerodynamics

- Ran [KAAN](#) store-separation aerodynamic studies end to end and reported the results: geometry preparation, meshing, solver execution, 6-DOF trajectory analysis, post-processing; observed scaled-store wind-tunnel tests across multiple configurations.
- Delivered a first-principles flight-dynamics analysis of the KAAN fighter: derived the 6-DOF Newton–Euler equations of motion, stability derivatives and spin behavior and documented how the aerodynamic database and surrogate models are built.
- Delivered a technical report on T129 ATAK rotor-blade manufacturing and balancing to evaluate production of AKSUNGUR UAV propeller blades (master tooling, tolerances, CG control), including gram-scale tip trim-mass corrections and their moment-arm effect on balance and vibration.
- Documented the design-to-production data chain (composite manufacturing, Teamcenter routing, NX/Fibersim ply data, NCR/MRB) across the AKSUNGUR, ANKA-3, Super Şimşek and AYAZ programs, analyzing process consistency

**Research Intern** · Computational Nanomechanics, New Jersey Institute of Technology Gor Laboratory Summer 2025  
– Ran Grand Canonical Monte Carlo simulations of helium confined in zirconium nanovoids at 573 K and 973 K (Mezcal), establishing thermodynamic baselines against irradiation literature; built Python/Jupyter workflows automating runs, equilibrium assessment and density/pressure-vs- $\mu$  analysis.

## PROJECTS & ADDITIONAL EXPERIENCE

**Founder & Electrical Technician** · Hi Tech Electrical Contractors Inc., NJ 2020 – Present  
Electrician since 2020; incorporated the company in 2023 and operate it while studying full time; perform panel upgrades, branch-circuit wiring, NEC-compliant installations, and fault diagnostics. Home Depot Pro Referral partner with **22 verified five-star reviews** (5.0/5.0). [\[Link\]](#).

**Electric Vehicle Conversion** · Bergen Community College STEM Research Center 2024 – 2025  
Converted a 1998 Chevrolet to an electric drivetrain; developed a CAN Bus sensor monitoring and fault-diagnostic system.

**Rutgers Propulsion Lab (AIAA)** · Member 2026 – Present  
Contributed CAD models for structural components for student rocket development; investigated paraffin-based hybrid rocket fuels.

## SERVICE & VOLUNTEERING

**Environmental Service Volunteer** · USDA 3SP Scholarship Program Bergen Community College  
Completed a semester of environmental service, including campus tree planting and invasive-species removal; received a \$3,000 USDA 3SP scholarship.

## INTERESTS

Computational Aerospace | Biomimetic Flight Mechanics | Swimming | Fitness | Aviation History | Road Trips